

30264 - Public Finance

BIEM
Spring 2026



Social Insurance
Gruber - Chapter 12

1 What Is Insurance and Why Do Individuals Value It?

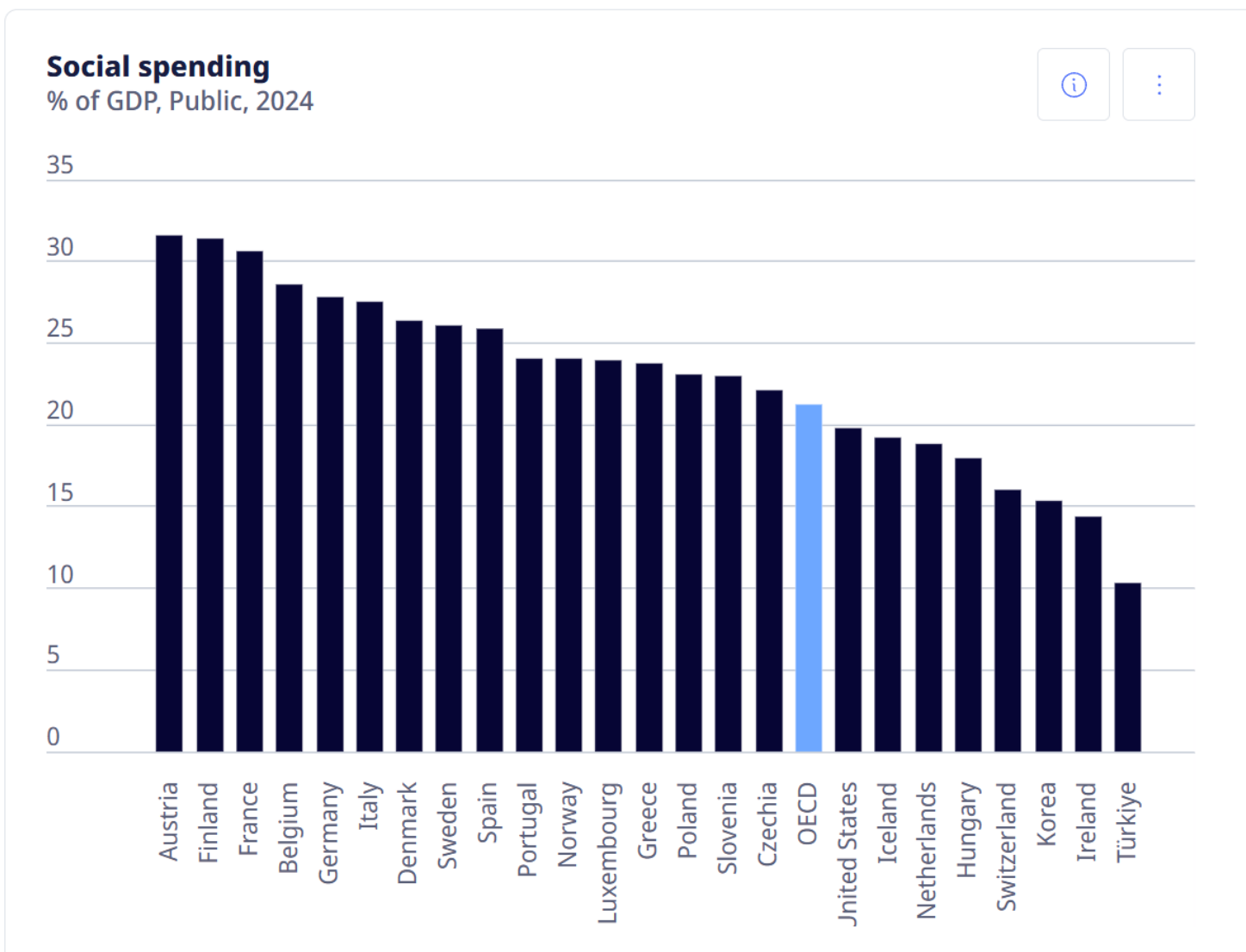
2 Why Have Social Insurance? Asymmetric Information and Adverse Selection

3 The Problem with Insurance: Moral Hazard

Social Insurance Programs (in bold what we'll cover)

- **Social Security (Pensions)**
- Unemployment Insurance
- Disability Insurance
- Worker's compensation
- **Health Insurance**
- **Social Exclusion/Anti-poverty measures**
- (Housing)

Social spending as % of GDP for 2024, OECD countries



EU vs US model of social insurance

1. «European Model»

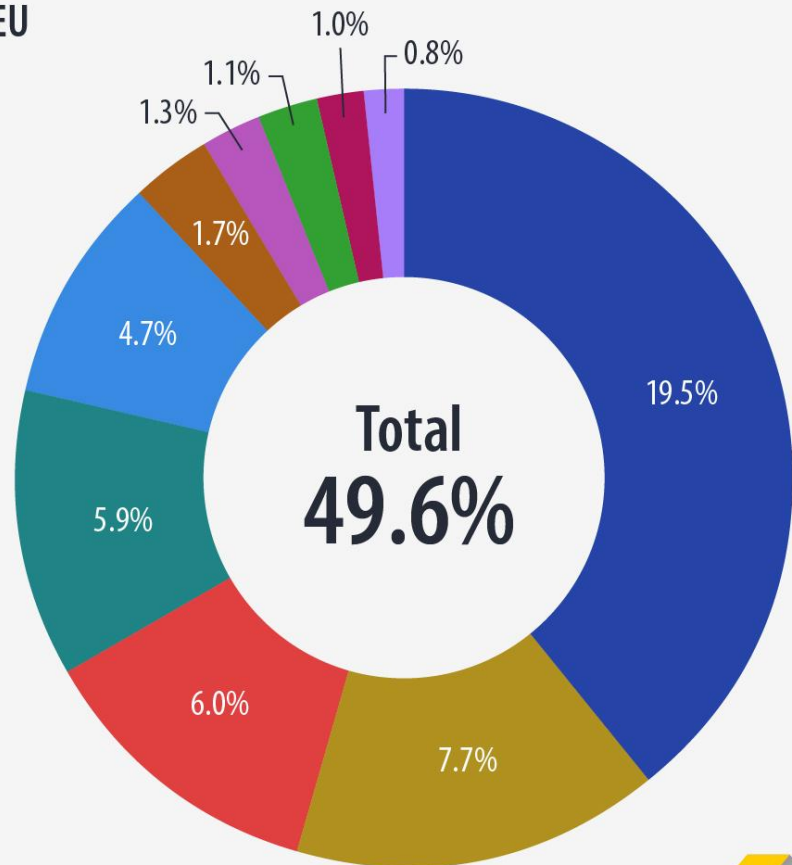
- Crucial role of the government in the direct provision of social insurance

2. «Private Model» → private expenditure (+ fiscal incentives)

- **Household private direct expenditure**
- **Private Insurance**

General government expenditure by function in the EU in 2022 (% of GDP)

- Social protection
- Health
- General public services
- Economic affairs
- Education
- Public order & safety
- Defence
- Recreation, culture & religion
- Housing & community amenities
- Environmental protection



Data is provisional

eurostat 

https://ec.europa.eu/eurostat/statistics-explained/index.php/Government_expenditure_by_function_-_COFOG#General_overview

What is insurance and why do individuals value it?

→ Insurance is a **promise** to make some **payment** in case of a particular **negative event**, in exchange for a payment, called a **premium**

- **States of the world:** The set of outcomes that are possible in an uncertain future.
- **Insurance premiums:** Money that is paid to an insurer so that an individual will be insured against adverse events.

→ Insurance is valuable because it helps individuals **level consumption across states of the world**.

In the next slide, a quick recap of the **expected utility model**: the utility of a “lottery” is the weighted sum of utilities across states of the world, where the weights are the probabilities of each state occurring.

Suppose an adverse event occurs with probability p . The expected utility is

$$EU = (1 - p) \times U(\text{cons. with no adverse event}) + p \times U(\text{cons. with adverse event})$$

No insurance scenario:

w = wealth if no accident occurs

$w - d$ = wealth if accident occurs

According to the expected utility model, the expected utility will be:

$$EU = (1-p)U(w) + pU(w-d)$$

Instead, consider the possibility of purchasing an **insurance contract**:

b = reimbursement

m = unitary premium ($0 < m < 1$) per dollar of reimbursement

→ total premium is mb

Upon purchasing this contract, the lottery faced by the individual has the following payoffs:

$w - mb$ = wealth if no accident occurs

$w - mb - d + b$ = wealth if accident occurs

So the expected utility is

$$EU = (1-p)U(w-mb) + pU(w-mb-d+b)$$

?What do you prefer?

If the premium is **actuarially fair** and the contract provides **full insurance**, then **risk-averse** individuals will always purchase that contract (it maximizes their expected utility).

Actuarially fair: $m=p$, the premium reflects the probability

Full insurance: $b=d$, the damage is fully reimbursed

Risk-averse: concave utility function, i.e. diminishing marginal utility → these individuals do not like risk, they prefer certainty to lotteries having the same expected value (they are also willing to pay to avoid risk)

Intuition: risk-averse people prefer giving up some of their money to avoid risk.

Insurance is valuable because it helps individuals **level consumption across states of the world**.

Numerical Example

- $p=1\%$ chance that Sam gets hit by a car, resulting in \$30,000 in medical expenses.
- Insurance costs m for each dollar of coverage.
 - If Sam buys $\$b$ of coverage, his premium is $\$mb$.
- To analyze Sam's choice, assume $U = \sqrt{c}$, and premiums are actuarially fair.

Purchase	Hit?	c	$\sqrt{c}$	Expected Utility
No insurance	Yes	0	0	$0.01 \times 0 + 0.99 \times 173.2$ $= \mathbf{171.5}$
	No	30,000	173.2	
Full insurance (\$300)	Yes	29,700	172.34	$0.99 \times 172.34 + 0.01 \times 172.34$ $= \mathbf{172.34}$
	No	29,700	172.34	
Partial insurance (\$150)	Yes	14,850	121.86	$0.99 \times 172.77 + 0.01 \times 121.86$ $= \mathbf{172.26}$
	No	29,850	172.77	

The Role of Risk Aversion

- **Risk aversion:** The extent to which individuals are willing to bear risk.
 - Risk-averse people may still want to buy some insurance even if it is not actuarially fair.
 - People may differ in their risk aversion, and if insurance premiums are extremely unfair, then only the most risk averse will want it.

We said that risk-averse individual like buying insurance (they prefer full insurance when the premium is actuarially fair), so....

...why is insurance a public finance topic? What is wrong in this market?

INFORMATIONAL ASYMMETRIES

- The equilibrium reached by private agents could be not Pareto efficient if there are some informational asymmetries in the insurance market

→ **Ex-ante asymmetry (adverse selection)**

(Ex-post asymmetry (moral hazard) can be caused by insurance itself...more on this later...)

Why Have Social Insurance? Asymmetric Information and Adverse Selection

Why should the government provide insurance?

- Information asymmetry can lead to a key market failure: adverse selection.
 - **Information asymmetry:** The difference in information that is available to sellers and to purchasers in a market.
- In the insurance market, buyers may know more about their insurable risks than the seller (insurer) does.
- **Adverse selection:** The fact that the insured individuals know more about their risk level than does the insurer might cause those most likely to have the adverse outcome to select insurance, leading insurers to lose money if they offer insurance.

Adverse Selection Example

- Two kinds of people:
 - People texting while driving have a 5% chance of being in a car accident (half the population).
 - Careful people have a 0.5% chance (half the population).
- If the insurance companies **knows each person's type (full information)**, it can charge them separate prices
- If the insurance company **doesn't know their type**, it could try charging a price that is **fair on average (pooling equilibrium)**, or try charging **separate prices (separating equilibrium)**

Insurer Breaks Even with **Full Information Pricing**

- What happens if the insurance company could charge each type their actuarially fair price?
 - Charge careless people \$1,500.
 - Charge careful people \$150.
 - Earn \$150,000 per 100 careless people, pay out \$150,000.
 - Earn \$15,000 per 100 careful people, pay out \$15,000.

Asymmetric Information Pricing: **Average price**

- What if the insurance company tries to charge average price?
 - Average price: $\$825 = 0.5 * 1500 + 0.5 * 150$
 - Insurance is a great deal for careless people, so they buy it, pay \$825.
 - Careful people decline it.
 - Earn \$82,500 per 100 careless people, pay out \$150,000. **Lose \$67,500.**
 - Earn nothing from careful people.

Asymmetric Information Pricing: **Separate prices**

- What if the insurance tries to charge different prices but cannot tell who is careless?
 - Careless people pretend to be careful, pay \$150.
 - Careful people pay \$150.
 - Earn \$15,000 per 100 careless people, pay out \$150,000. **Lose \$135,000.**
 - Earn \$15,000 per 100 careful people, pay out \$15,000.

Adverse Selection Example

Information	Pricing Approach	Premium per Careless (100 people)	Premium per Careful (100 people)	Total Premiums Paid	Total Benefits Paid Out	Net Profits to Insurers
Full	Separate	\$1,500	\$150	\$165,000 (100 x \$1,500 + 100 x \$150)	\$165,000	0
Asymmetric	Separate	\$1,500	\$150	\$30,000 (0 x \$1,500 + 200 x \$150)	\$165,000	-\$135,000
Asymmetric	Average	\$825	\$825	\$82,500 (100 x \$825 + 0 x \$825)	\$150,000	-\$67,500

The Problem of Adverse Selection

- **Adverse selection:** The fact that insured individuals know more about their risk level than does the insurer might cause those most likely to have the adverse outcome to select insurance, leading insurers to lose money if they offer insurance.
- Selling to both requires that low-risk people subsidize high-risk people.
- Low-risk people may not want to do this.
- Sometimes, only high-risk people end up with insurance.

Does Asymmetric Information Necessarily Lead to Market Failure?

If low-risk people have a high enough risk premium (i.e. they really dislike risk!), they will subsidize high-risk people in a pooling equilibrium.

- **Risk premium:** The amount that risk-averse individuals will pay for insurance above and beyond the actuarially fair price.
- **Pooling equilibrium:** A market equilibrium in which all types of people buy full insurance even though it is not fairly priced to all individuals.
- **Separating equilibrium:** A market equilibrium in which different types of people buy different kinds of insurance products designed to reveal their true types.

Does Asymmetric Information Necessarily Lead to Market Failure?

Alternatively, the insurance company could offer two alternative contracts

Contract A: fair premium for high risk people + full insurance

Contract B: fair premium for low risk people + partial insurance

Aim → type revelation

High risk buy contract A

Low risk buy contract B

- **Separating equilibrium:** A market equilibrium in which different types of people buy different kinds of insurance products designed to reveal their true types.

→ **market failure!!** Low risk would like to buy full insurance, not offered to them!

How Does the Government Address Adverse Selection?

Adverse selection leads to market failure, since healthy people may not be willing to buy insurance.

- The government can address adverse selection and improve market efficiency in a number of ways . . .
- but they all involve redistribution from the healthy to the sick, which may be unpopular.

Other Reasons for Government Intervention in Insurance Markets

- **Externalities:** Vaccines have positive spillovers; car crashes negative ones.
- **Administrative costs:** Government-run Medicare has much lower administrative costs than private insurance.
- **Redistribution:** Governments may want to redistribute from healthy to sick.
- **Paternalism:** Governments may feel that people would choose to buy too little insurance for themselves.

Social Insurance versus Self-Insurance: How Much Consumption Smoothing?

Even if private insurance markets do not function well, people may still be able to insure with self-insurance.

- **Self-insurance:** The private means of smoothing consumption over adverse events, such as through one's own savings, the labor supply of family members, or borrowing from friends.

Example: Unemployment Insurance

People can insure against unemployment in many ways:

- They can draw on their own savings.
- They can borrow, either in *collateralized* forms (such as borrowing against the equity they have in their homes) or in *uncollateralized* forms (such as on their credit card).
- Other family members can increase their labor earnings.
- They can receive transfers from their extended family, friends, or local organizations.

Example: Unemployment Insurance - effectiveness

How much self-insurance is in place determines how effective social insurance is.

- For an individual with no self-insurance, each dollar of unemployment insurance (UI) directly reduces the decline in consumption caused by unemployment.
 - No crowding out
- For an individual with complete self-insurance, each dollar of UI replaces a dollar of self-insurance.
 - Complete crowding out
- For an individual with partial self-insurance, each dollar is split between smoothing consumption and reducing self-insurance.
 - Partial crowding out

Lessons for Consumption-Smoothing Role of Social Insurance

The importance of social insurance for consumption smoothing will depend on two factors:

1. *Predictability of the event*: It is easier for people to self-insure against a predictable event, such as increasing their savings. More predictable risks reduce the benefits of providing social insurance.
2. *Cost of the event*: It is more difficult to self-insure against high-cost events, such as becoming injured and unable to work. Costly risks increase the benefits of providing social insurance.

The Problem with Insurance: Moral Hazard

The cost of insurance is moral hazard.

- **Moral hazard:** Adverse actions taken by individuals or producers in response to insurance against adverse outcomes.
- The existence of moral hazard means that it may not be optimal for the government to provide the full insurance that is demanded by risk-averse consumers.

What determines moral hazard?

- How easy it is to observe whether the adverse event has happened.
- How easy it is to change behavior in order to establish the adverse event.

The Problem with Insurance:

Moral Hazard in social insurance

In examining the effects of social insurance, four types of moral hazard play a particularly important role.

- Reduced precaution against entering the adverse state.
 - Because you have medical insurance, you reduce efforts to protect your health.
- Increased odds of entering the adverse state.
 - Because you have workers' compensation, you are more likely to claim that you were injured on the job.
- Increased expenditures when in the adverse state.
 - Because you have medical insurance, you use more medical care.
- Supplier responses to insurance against the adverse state.
 - Because you have workers' compensation, firms aren't as careful about protecting you against workplace accidents.

The Consequences of Moral Hazard

Moral hazard is costly for two reasons.

- The adverse behavior encouraged by insurance lowers social efficiency; for example, workers' compensation reduces the provision of a socially efficient labor supply.
- When social insurance encourages adverse events, which raise the cost of the social insurance program, it increases taxes and lowers social efficiency further.

Putting It All Together: Optimal Social Insurance

- Optimal social insurance systems should partially, but not completely, insure individuals against adverse events.
- The benefit of social insurance is the amount of consumption smoothing provided by social insurance programs.
- The cost of social insurance is the moral hazard caused by insuring against adverse events.

Conclusion

- Asymmetric information in insurance markets has two important implications:
 - It can cause *adverse selection*
 - It can cause *moral hazard*
- The ironic feature of asymmetric information is therefore that it simultaneously motivates *and* undercuts the rationale for government intervention through social insurance.

Exercises

1) Adverse selection arises in insurance markets because:

- a) Insurers have better information than consumers
- b) Insurance causes individuals to change behavior
- c) Individuals know more about their risk than insurers
- d) Government regulation distorts prices

2) Which of the following is correct about pooling equilibria in insurance contexts?

- a) High-risk and low-risk people pay the same premium
- b) High-risk people will purchase partial coverage contracts
- c) Low-risk people maximize their utility
- d) The insurance company makes positive profits

3) In a separating equilibrium:

- a) Low-risk individuals subsidize high-risk individuals
- b) Individuals reveal their type by choosing different contracts
- c) Everyone receives full insurance
- d) The market always achieves efficiency

4) Which of the following is an example of self-insurance?

- a) Buying health insurance from the government
- b) Saving money to prepare for unemployment
- c) Receiving unemployment benefits
- d) Mandatory social security contributions

5) There are two groups of people in city A: bad drivers and good drivers. The probability of incurring in an accident is respectively 20% for bad drivers and 5% for good drivers. The insurance company cannot discriminate between the two groups of people and offers a unique contract with a unitary premium of 0,15\$. This implies that the proportion of good drivers in the population is

- a) One third
- b) Two thirds
- c) Uncertain
- d) One half

TRUE/FALSE (please provide a justification)

- 1) Insurance is valuable for risk-averse individuals because it increases expected income.
- 2) Adverse selection arises because insurance coverage changes individuals' behavior after the contract is signed.
- 3) In a pooling equilibrium, low-risk individuals may subsidize high-risk individuals.

Exercise 1

Mr. Rossi has a 2% probability of suffering from sinusitis. In case of illness, in the past he has always been treated with pharmaceutical therapy, with a total cost (damage) of 10 EUR. Given this information, an insurance company offers him an insurance policy with an actuarially fair premium.

- a) What is the unit premium charged by the insurance company?
- b) At this premium, will Mr. Rossi purchase partial or full coverage?
- c) What is the total premium paid by Mr. Rossi to the insurance company?

Once the insurance coverage is purchased, Mr. Rossi's trusted clinic informs him about the possibility of treating sinusitis more effectively and without side effects through thermal water inhalations, costing 250 EUR. The clinic receives retrospective reimbursement from the insurance company, without the insurer being able to observe which treatment is most appropriate for the specific case.

- d) Explain in words and quantify numerically the problem that arises in this case.
- e) What strategy can the insurance company adopt to limit this problem?
- f) Answer again questions (d) and (e) in the case in which the insurance company acquires the medical clinic.

Exercise 2

The population of Gotham City consists of 2000 people, 200 criminals and 1800 virtuous citizens. The probability of being injured is 30% for criminals and 10% for virtuous citizens. The damage that a person would suffer from an injury is 300\$ (independently from their type).

- a) Please compute the premium (in dollars) that criminals and virtuous citizens would be asked to pay by risk-neutral insurance company in case their type was observable and they were offered full coverage.
- b) Please compute the profits of the insurance company in the above case.
- c) What if the insurance company was unable to distinguish between the two types of citizens? What premium (in dollars) would they be asked to pay (assume full coverage as above)?
- d) How do you call such an equilibrium described in part c)? Is it always achievable? Please briefly discuss.